



AI Driven SQA: Manual & Automation Testing

Assignment 04

Module: 04

**Topic: Introduction to Programming Language
with AI (JavaScript + Node.js).**

Submitted by,

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Q1. JavaScript Coding Practice.

01. Check whether a number is even or odd.

Explanation: This program is developed using JavaScript in the Node.js environment to check whether a user-entered number is even or odd. The program uses the built-in readline module of Node.js to take input from the user through the terminal.

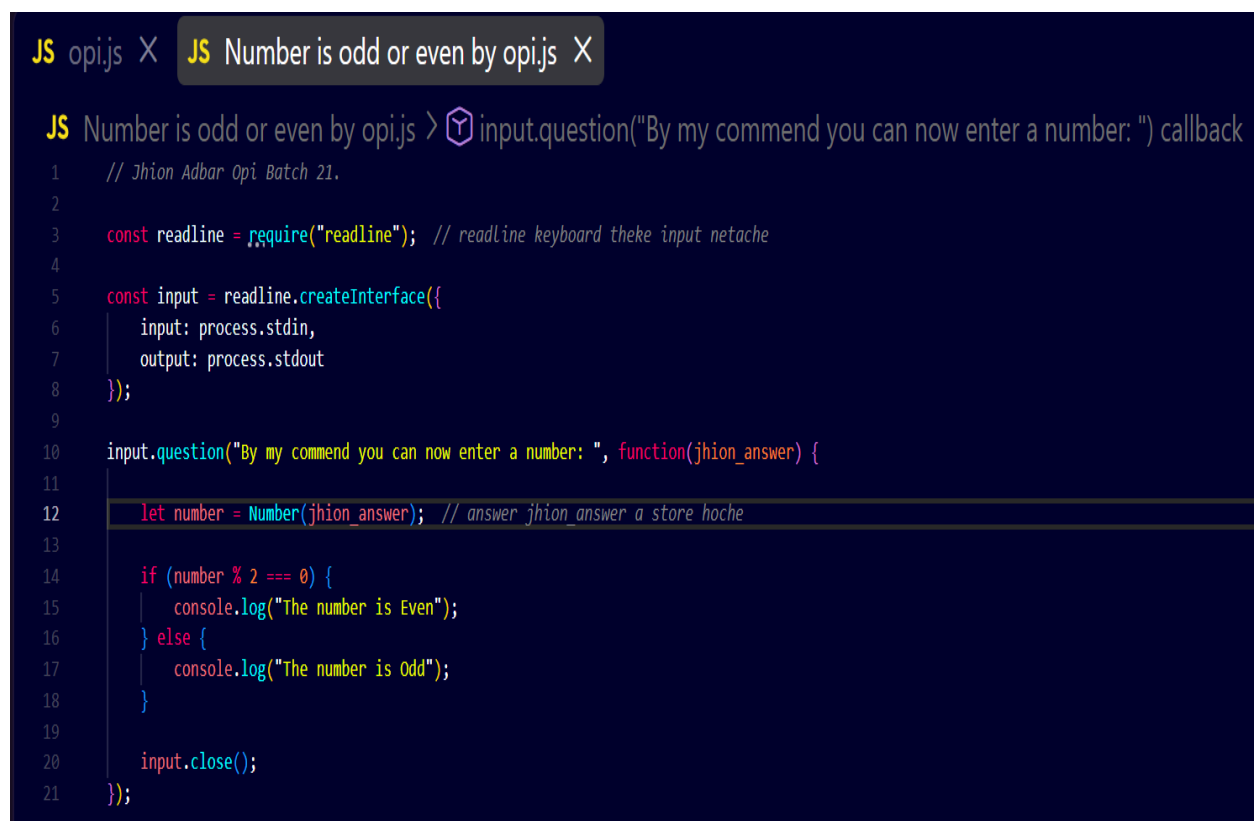
First, the program creates an input interface using `readline.createInterface()` to receive data from the keyboard. Then asked to enter a number. Since user input is received as a string, the `Number()` function is used to convert the input into a numerical value.

After that, an if-else conditional statement along with the modulus operator (%) to determine the type of number. If the remainder after dividing the number by 2 is 0, the number is even or it is considered odd.

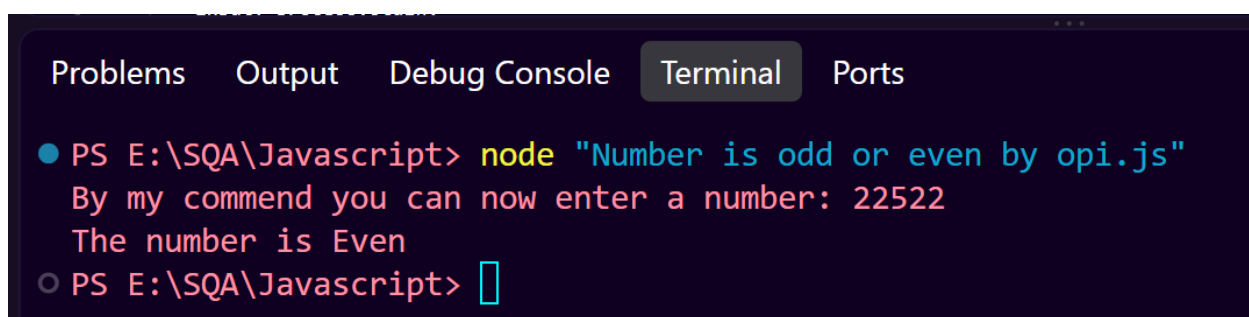
Finally, the program displays the result on the terminal and closes the input interface after completing the operation.

Source code: <https://drive.google.com/file/d/1IHtROQaQ59VgLcldPHcDAjAkEFAShgk/view?usp=sharing>

Output:



```
JS Number is odd or even by opi.js X
JS Number is odd or even by opi.js > input.question("By my commend you can now enter a number:") callback
1 // Jhion Adbar Opi Batch 21.
2
3 const readline = require("readline"); // readline keyboard theke input netache
4
5 const input = readline.createInterface({
6   input: process.stdin,
7   output: process.stdout
8 });
9
10 input.question("By my commend you can now enter a number: ", function(jhion_answer) {
11
12   let number = Number(jhion_answer); // answer jhion_answer a store hoche
13
14   if (number % 2 === 0) {
15     console.log("The number is Even");
16   } else {
17     console.log("The number is Odd");
18   }
19
20   input.close();
21 });
```



```
Problems Output Debug Console Terminal Ports
● PS E:\SQA\Javascript> node "Number is odd or even by opi.js"
  By my commend you can now enter a number: 22522
  The number is Even
○ PS E:\SQA\Javascript> 
```

Jhion Adbar Opi

02. Find the largest of three numbers.

Explanation: This program finds the largest value among three numbers entered by the user. It uses the readline module in Node.js to collect input from the terminal. Since the entered values are received as strings, the Number() function converts them into numerical values. The program uses if-else if-else conditional statements to compare the three numbers. The logical AND operator (&&) ensures that a number is greater than or equal to the other two numbers before declaring it the largest. Finally, the program displays the largest number and closes the input interface.

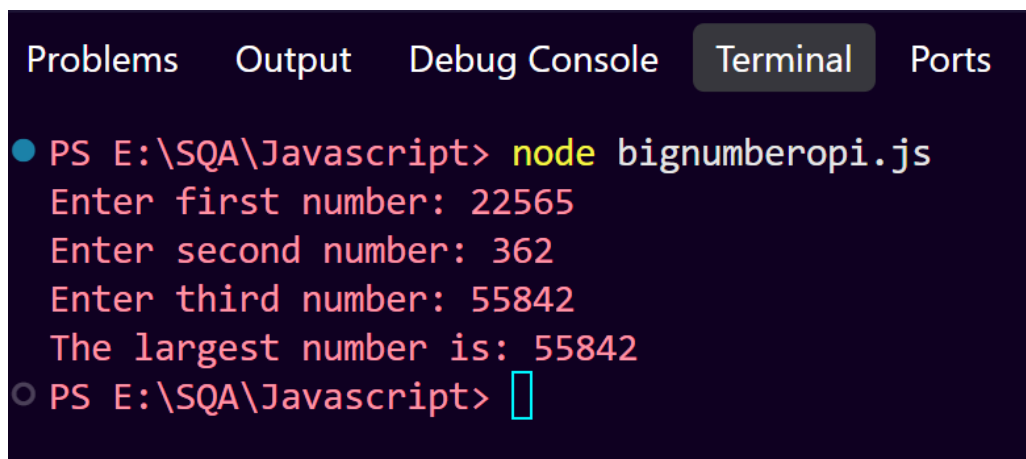
Source code: <https://drive.google.com/file/d/1VBKKdW0lC83ApKW2Q2DnVkWEuLxPOYGh/view?usp=sharing>

Output:



```
JS bignumberopi.js X JS Number is odd or even by opi.js

JS bignumberopi.js > ...
1 // Jhion Adbar Opi Batch 21
2 // Learge number
3
4 const readline = require("readline"); // readline keyboard theke input netache
5
6
7 const input = readline.createInterface({
8   input: process.stdin,
9   output: process.stdout
10 });
11
12 input.question("Enter first number: ", function(first) {
13   input.question("Enter second number: ", function(second) {
14     input.question("Enter third number: ", function(third) {
15       let num1 = Number(first);
16       let num2 = Number(second);
17       let num3 = Number(third);
18
19       if (num1 >= num2 && num1 >= num3) {
20         console.log("The largest number is: " + num1);
21       }
22       else if (num2 >= num1 && num2 >= num3) {
23         console.log("The largest number is: " + num2);
24       }
25       else {
26         console.log("The largest number is: " + num3);
27       }
28
29       input.close();
30     });
31   });
32 });
33
34
35
36
37
38
```



```
Problems Output Debug Console Terminal Ports

● PS E:\SQA\Javascript> node bignumberopi.js
Enter first number: 22565
Enter second number: 362
Enter third number: 55842
The largest number is: 55842
○ PS E:\SQA\Javascript> 
```

03. Reverse a string.

Explanation: This JavaScript program takes a string as input from the user using the ``readline`` module. The program then reverses the characters of the string using the ``split()``, ``reverse()``, and ``join()`` methods. Finally, it displays the reversed string on the console. This program demonstrates user input handling and basic string manipulation in JavaScript.

Source Code: https://drive.google.com/drive/folders/1Nk2kSCL2AB40Uu00OxxVwIM4PCV8_GTp?usp=drive_link

Output:

```
JS reverse_a_string_by_opi.js X JS bignumberopi.js X

JS reverse_a_string_by_opi.js > ...
1 // Reverse a string
2 // Jhion Adbar Opi Batch 21
3
4 const readline = require("readline");
5
6 const input = readline.createInterface({
7   input: process.stdin,
8   output: process.stdout
9 });
10
11 input.question("Enter a string: ", function(text) {
12
13   let reversed = text.split("").reverse().join("");
14
15   console.log("Reversed string: " + reversed);
16
17   input.close();
18 });
```

```
Problems Output Debug Console Terminal Ports

● PS E:\SQA\Javascript> node reverse_a_string_by_opi.js
Enter a string: jhion_adbar_opi
Reversed string: ipo_rabda_noihj
● PS E:\SQA\Javascript> node reverse_a_string_by_opi.js
Enter a string: Jhion_Adbar_OPI_121
Reversed string: 121_IPO_rabda_noihJ
● PS E:\SQA\Javascript> node reverse_a_string_by_opi.js
Enter a string: Jhion_THE_GREAT
Reversed string: TAERG_EHT_noihJ
○ PS E:\SQA\Javascript> █
```

04. Count vowels in a string.

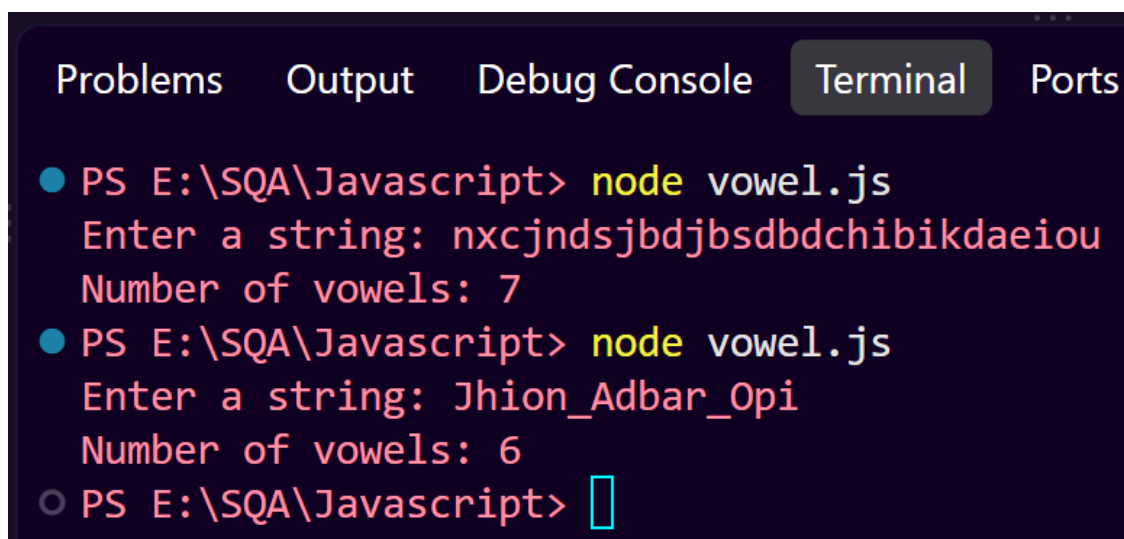
Explanation: This JavaScript program accepts a string as input from the user and iterates through each character using a loop. It checks whether the character is a vowel (a, e, i, o, u) using the `includes()` method and increments a counter when a vowel is found. Finally, the program displays the total number of vowels in the string.

Source Code: https://drive.google.com/drive/folders/1qzJJzVASNT_noyRpBN73KvPHxqDgZS8i?usp=drive_link

Output:



```
JS vowel.js X JS reverse_a_string_by_c
JS vowel.js > ...
1 // Count vowels in a string
2 // Jhion Adbar Opi Batch 21
3
4 const readline = require("readline");
5
6 const input = readline.createInterface({
7   input: process.stdin,
8   output: process.stdout
9 });
10
11 input.question("Enter a string: ", function(text) {
12
13   let count = 0;
14   let vowels = "aeiouAEIOU";
15
16   for (let i = 0; i < text.length; i++) {
17     if (vowels.includes(text[i])) {
18       count++;
19     }
20   }
21
22   console.log("Number of vowels: " + count);
23
24   input.close();
25 });
```



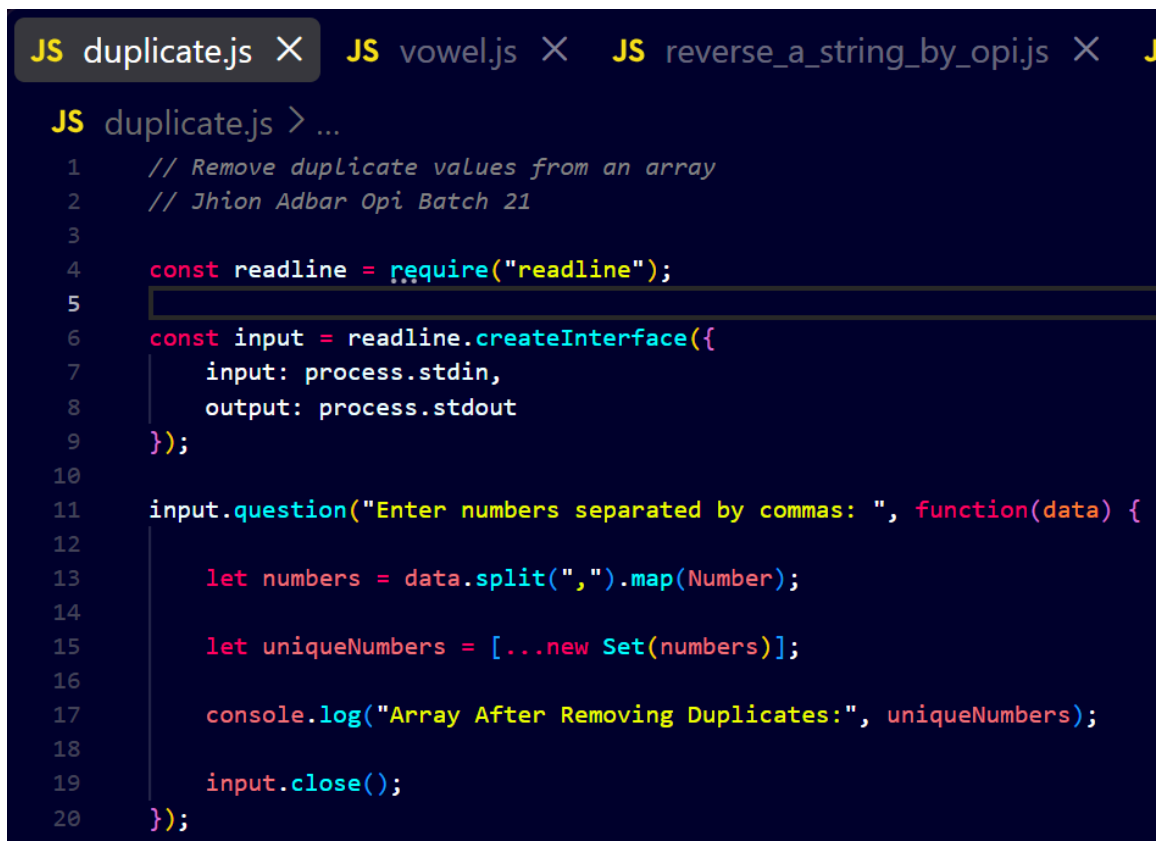
```
Problems Output Debug Console Terminal Ports
● PS E:\SQA\Javascript> node vowel.js
Enter a string: nxcjndsjbdjbsdbdchibikdaeiou
Number of vowels: 7
● PS E:\SQA\Javascript> node vowel.js
Enter a string: Jhion_Adbar_Opi
Number of vowels: 6
○ PS E:\SQA\Javascript> 
```

05. Remove duplicate values from an array.

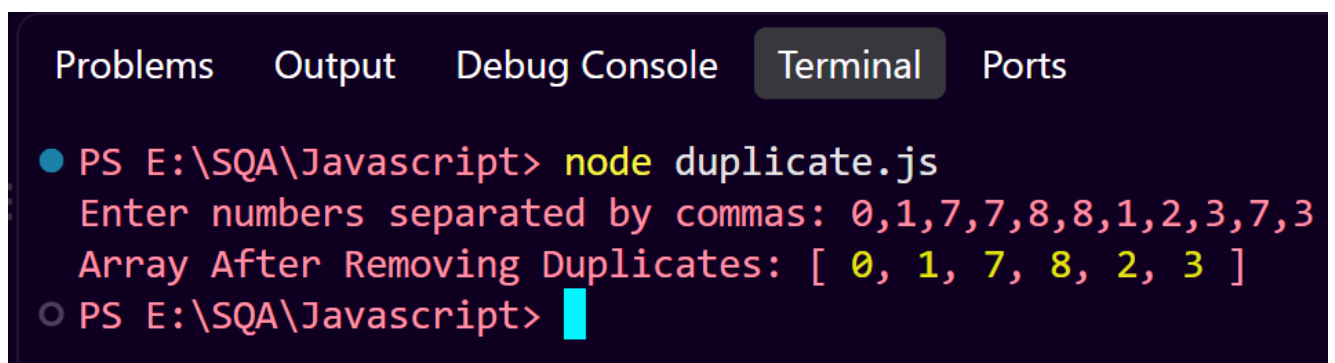
Explanation: This JavaScript program removes duplicate values from an array using the `Set` object. The array elements are converted into a `Set`, which automatically stores only unique values. The unique values are then converted back into an array and displayed as the output.

Source Code: https://drive.google.com/drive/folders/1k0FPPBaiNQDb882n8-9pwxEX3uzNaLB2?usp=drive_link

Output:



```
JS duplicate.js X JS vowel.js X JS reverse_a_string_by_opi.js X JS
JS duplicate.js > ...
1 // Remove duplicate values from an array
2 // Jhion Adbar Opi Batch 21
3
4 const readline = require("readline");
5
6 const input = readline.createInterface({
7   input: process.stdin,
8   output: process.stdout
9 });
10
11 input.question("Enter numbers separated by commas: ", function(data) {
12
13   let numbers = data.split(",").map(Number);
14
15   let uniqueNumbers = [...new Set(numbers)];
16
17   console.log("Array After Removing Duplicates:", uniqueNumbers);
18
19   input.close();
20 });
```



```
Problems Output Debug Console Terminal Ports
● PS E:\SQA\Javascript> node duplicate.js
Enter numbers separated by commas: 0,1,7,7,8,8,1,2,3,7,3
Array After Removing Duplicates: [ 0, 1, 7, 8, 2, 3 ]
○ PS E:\SQA\Javascript>
```